CYBERNETICA

Internship Program 2026
Test assignment for Junior Security Engineer

Introduction

The Digital Identity Technologies Department uses cryptography (encryption, signatures, …), security protocols (zero-knowledge proofs, multi-party computation, …), and technology (WebAuthn, smart-cards, secure elements, biometrics, …) to build products like:

1. Smart-ID eID solution used in Estonia, Latvia and Lithuania
2. SplitKey for private key protection (inside Smart-ID and other systems)
3. Secure online voting system used in Estonia
4. CDOC2 solution for document encryption in DigiDoc4
5. Upcoming EU digital identity wallets and components

If these products sound fascinating, raise questions, or if you want to see how university building blocks create real-life, future-proof technologies for secure digital societies, this is your chance.

Requirements

To solve real tasks during the summer, you must already have the following skills:

1. Read and write code (Java, Rust, Python) to solve problems.
2. Write technical English documentation and create diagrams (UML, BPMN, threat modelling).
3. Understand cryptography building blocks (encryption, signatures; e.g., , "MTAT.07.017 Applied Cryptography" at Tartu University).
   We will teach other skills on the spot.

Application to eID department.

Sounds interesting and you would like to give it a try?

Great! As a next step we would like you to write to us a short letter (maximum 1 A4), that explains your interest and your motivation. This letter should somehow differentiate yourself from the rest of applicants. You could highlight your existing knowledge from the university curriculum, you could point out your existing development experience, you could explain why information security topic is close to your heart and what kind of technology you think is necessary to use in a future society and in products.

For inspiration, we can give you some topics here:

1. Why is phishing and fraud problem inherently very difficult problem to solve and what kind of technical security measures Estonia has deployed so far. Are they effective? Can you make them better?
2. Do you think that non-human identities (NHIs i.e, servers, processes, AI agents) are any different from human identities? What underlying technologies are used to secure NHIs that cannot be used to secure ordinary government-issued identities?
3. What security issues have you encountered in your real-life experiences so far and what you think, what kind of technology could have been useful to prevent or mitigate the outcome?
4. ...

P.S.

Using AI to write the letter is permitted, as we use these tools internally. However, disclose the tools and prompts used, and explain how you verified the results. Detail your manual contributions, including trial and errors. Include source code if applicable. Finally, be prepared to argue and defend every detail in the interview.